

CELL CYCLE & CELL DIVISION

♥ "From one cell to many - the journey of life continues." ♥

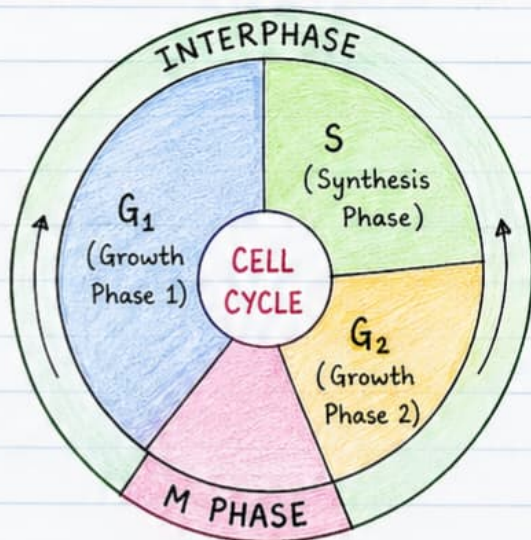
INTRODUCTION

- All living organisms originate from pre-existing cells.
- Cells increase in number through cell division.
- Before cell division, cell grows and duplicates its DNA. This entire sequence is called **Cell Cycle**.

CELL CYCLE

- The cell cycle is the sequence of events that occurs in a cell from one division to the next.
- It consists of two major phases :

1. **Interphase**
2. **M Phase (Mitotic Phase)**



INTERPHASE

- It is the longest phase of cell cycle.
- Cell prepares itself for division.
- It is further divided into three sub-phases :
 1. G_1 Phase (Gap 1)
 2. S Phase (Synthesis)
 3. G_2 Phase (Gap 2)

1. G_1 PHASE (Gap 1)

- Cell grows in size.
- Synthesis of RNA, proteins & organelles.
- Metabolically most active phase.
- Duration varies in different cells.



3. G_2 PHASE (Gap 2)

- Further growth of cell.
- Synthesis of proteins required for cell division.
- Energy is stored in the form of ATP.
- Cell checks for any DNA damage before entering M phase.



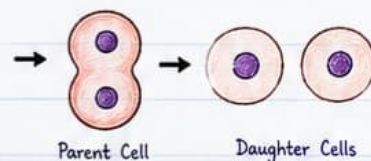
2. S PHASE (Synthesis Phase)

- DNA replication takes place.
- Each chromosome duplicates to form two sister chromatids.
- Histone proteins are synthesized.
- Amount of DNA becomes double.



M PHASE (Mitotic Phase)

- It is the phase of actual cell division.
- It includes two processes :
 1. Karyokinesis (Nuclear Division)
 2. Cytokinesis (Cytoplasmic Division)



Key Point : Cell Cycle → Interphase (G_1 , S, G_2) + M Phase (Karyokinesis + Cytokinesis)

MITOSIS (KARYOKINESIS)

Remember!

In Mitosis
Chromosome
Number
Remains Same
($2n \rightarrow 2n$)

Mitosis is the nuclear division in which one parent nucleus divides to form two daughter nuclei having identical genetic material.

STAGES OF MITOSIS

Mitosis consists of four sequential stages:

Prophase → Metaphase → Anaphase → Telophase

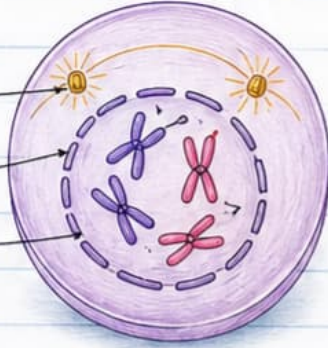
followed by Cytokinesis.

Where it occurs?

In somatic cells for growth, repair, and asexual reproduction.

1. PROPHASE

Centrosome (with centriole)
Disappearing nuclear envelope
Condensed chromosomes (each with two sister chromatids)



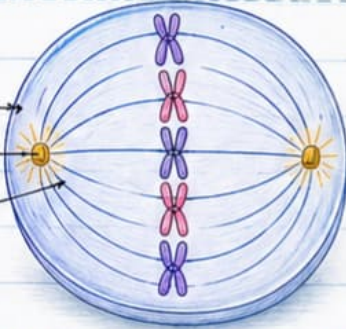
- Chromatin condenses to form visible chromosomes.
- Each chromosome has two sister chromatids joined at centromere.
- Nuclear envelope and nucleolus disappear.
- Centrosomes move to opposite poles.
- Spindle fibres start forming.



Key Point: $2n$ chromosomes are visible and each has two chromatids.

2. METAPHASE

Spindle fibre
Centromere
Chromosomes at equatorial plate



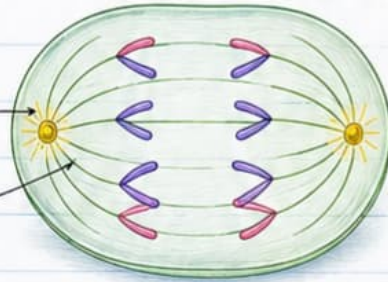
- Chromosomes align at the equatorial plate (middle of the cell).
- Spindle fibres attach to the kinetochores of centromere from both poles.
- This arrangement ensures equal distribution of chromosomes.



Remember: Metaphase is the best stage to study chromosomes (number, shape, size).

3. ANAPHASE

Daughter chromosomes (now separated)
Spindle fibres shortening



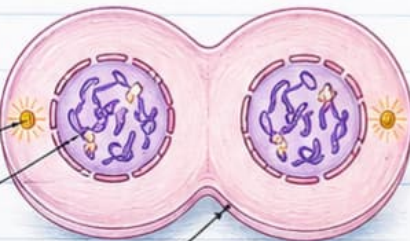
- Centromere splits into two.
- Sister chromatids separate and become daughter chromosomes.
- They are pulled towards opposite poles by spindle fibres.
- The cell elongates.



Key Point: Each pole receives an equal set of daughter chromosomes.

4. TELOPHASE + CYTOKINESIS

Re-forming nuclear envelope
Nucleolus reappears
Cleavage furrow



- Chromosomes reach poles and decondense to form chromatin.
- Nuclear envelope and nucleolus reappear.
- Spindle fibres disappear.
- Cytokinesis occurs — cytoplasm divides to form two daughter cells.

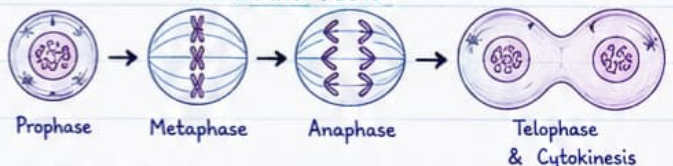


Result: Two genetically identical daughter cells are formed ($2n$ each).

SIGNIFICANCE OF MITOSIS

- Helps in growth of body.
- Repair and replacement of worn out cells.
- Regeneration in some organisms (e.g., Planaria).
- Asexual reproduction in unicellular organisms.

SUMMARY FLOW



One Parent Cell ($2n$) → Mitosis → Two Daughter Cells ($2n + 2n$)

MEIOSIS (REDUCTION DIVISION)

Meiosis is a type of cell division in which **diploid ($2n$)** cells divide twice to form **four haploid (n)** daughter cells.

Where it occurs?

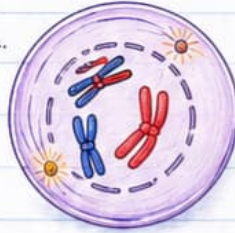
In germ cells (spermatogonia in testes and oogonia in ovaries) to produce gametes (sperm and ova).

STAGES OF MEIOSIS

MEIOSIS - I (REDUCTIONAL DIVISION)

1. PROPHASE I

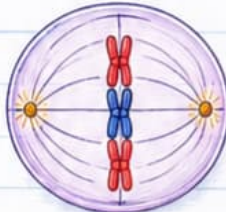
- Longest and most complex stage.
- Chromosomes condense.
- Homologous chromosomes pair up (synapsis) to form bivalents.
- Crossing over occurs between non-sister chromatids.
- Nuclear envelope and nucleolus disappear.
- Spindle fibres form.



Pairing of homologous chromosomes (bivalent) with crossing over

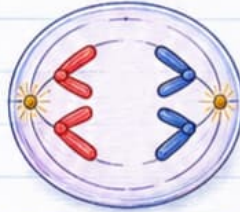
2. METAPHASE I

- Bivalents align at the equatorial plate.
- Orientation of each bivalent is random.



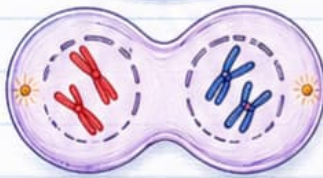
3. ANAPHASE I

- Homologous chromosomes separate and move towards opposite poles.
- Sister chromatids remain together.



4. TELOPHASE I & CYTOKINESIS

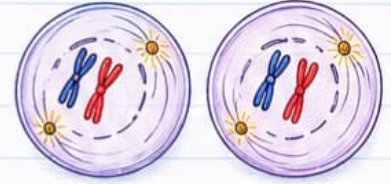
- Two haploid cells (n) are formed.
- Chromosomes are still in duplicated state.



MEIOSIS - II (EQUATIONAL DIVISION)

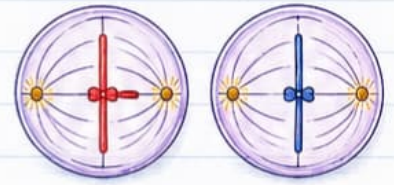
1. PROPHASE II

- Nuclear envelope (if present) breaks.
- Spindle fibres reappear.
- Chromosomes condense again.



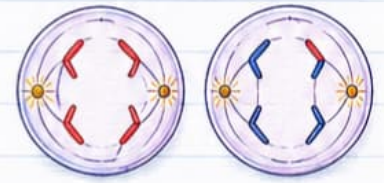
2. METAPHASE II

- Chromosomes align at equatorial plate of each cell individually.



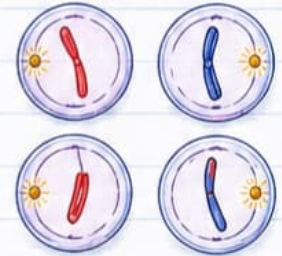
3. ANAPHASE II

- Centromere splits.
- Sister chromatids separate.
- They move towards opposite poles.



4. TELOPHASE II & CYTOKINESIS

- Nuclear envelope reforms.
- Cytokinesis occurs.
- Four haploid (n) daughter cells are formed.

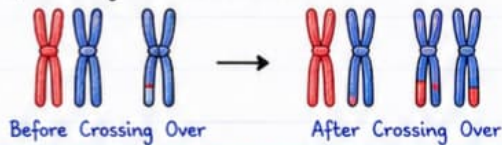


SIGNIFICANCE OF MEIOSIS

- Maintains chromosome number of the species by reducing it to half.
- Forms gametes for sexual reproduction.
- Introduces genetic variation through crossing over and independent assortment.

CROSSING OVER (in Prophase I)

Exchange of segments between non-sister chromatids of homologous chromosomes.



MITOSIS vs MEIOSIS

Feature	MITOSIS	MEIOSIS
Type of cell	Somatic cells	Germ cells (in gonads)
Number of divisions	One	Two (Meiosis I & II)
Number of daughter cells	Two	Four
Chromosome number in daughter cells	Same as parent cell ($2n \rightarrow 2n$)	Half of parent cell ($2n \rightarrow n$)
Synapsis of homologous chromosomes	Absent	Present (in Prophase I)
Crossing over	Absent	Present (in Prophase I)
Genetic variation	Daughter cells are genetically identical	Daughter cells are genetically different
Role	Growth, repair, replacement of cells	Formation of gametes and genetic variation

Quick Revision



Memory Tip

"Mitosis = Make Identical Things"

"Meiosis = Make Unique Gametes"



REGULATION OF CELL CYCLE



Cell cycle is tightly regulated to ensure accurate DNA replication and cell division.

Remember!

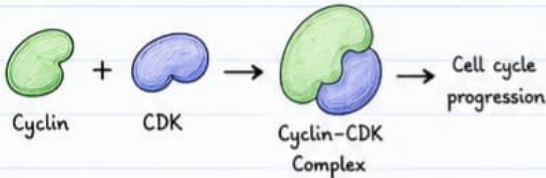
- ✓ Checkpoints act like "Quality Control".
- ✓ They ensure that each step is completed correctly before the next step begins.



1. CYCLINS AND CDKS

Cyclins are regulatory proteins whose levels rise and fall during the cell cycle.

- ★ They bind to Cyclin Dependent Kinases (CDKs).
- ★ Cyclin-CDK complex drives the cell cycle forward.
- ★ Different cyclins are active at different phases.



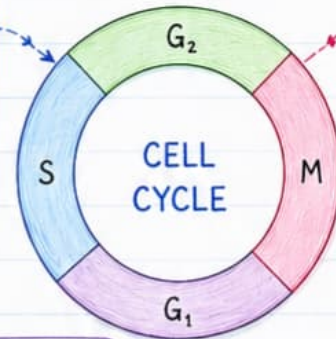
Cyclin-CDK complex = Engine that drives the cell cycle. ★

2. CHECKPOINTS

Checkpoints are control points that monitor the accuracy of events before the cell moves ahead.

G₂ CHECKPOINT

Checks if DNA replication is complete and if DNA is undamaged.



M CHECKPOINT

(Spindle assembly checkpoint)

Checks if all chromosomes are properly attached to spindle before anaphase.

G₁ CHECKPOINT

Checks cell size, nutrient availability, and DNA damage before DNA synthesis.

Analogy
Checkpoints are like traffic signals that allow the cell to proceed only when everything is OK! 😊

3. ROLE OF TUMOUR SUPPRESSOR GENES AND ONCOGENES

Tumour Suppressor Genes

- ★ Act as brakes for cell cycle.
- ★ They slow down or stop cell cycle when there is DNA damage.
- ★ Example: p53, Rb protein
- ★ Mutation → Loss of function → Uncontrolled cell division → Cancer

Tumour Suppressor Gene (Brake)



Slows down cell division

Oncogene (Accelerator)



Speeds up cell division

Oncogenes

- ★ Act as accelerators for cell cycle.
- ★ Promote cell division.
- ★ Mutation → Gain of function → Uncontrolled cell division → Cancer
- ★ Example: RAS, MYC

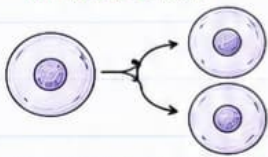


Balance between brakes and accelerators keeps cell division normal.
Imbalance → Cancer

4. CELL CYCLE AND CANCER

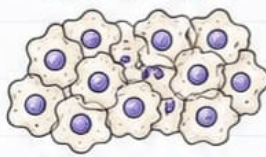
- ★ Cancer is caused by uncontrolled cell division.
- ★ Due to mutations in genes that regulate cell cycle.
- ★ These mutations may inactivate tumour suppressor genes or activate oncogenes.
- ★ Result → Checkpoints fail → Abnormal cell growth.

Normal Cell Division



Controlled & Limited

Cancer Cell Division



Uncontrolled & Unlimited

Key Fact

Cancer cells ignore checkpoints and never stop dividing!



5. SIGNIFICANCE OF CELL CYCLE

- ★ Growth and development of multicellular organisms.
- ★ Repair and replacement of worn out or dead cells.
- ★ Regeneration in some organisms (e.g., Planaria).
- ★ Maintains genetic stability by ensuring accurate DNA replication and distribution.



QUICK SUMMARY

- Cell Cycle = Interphase (G₁, S, G₂) + M Phase
- Mitosis = One division → 2 daughter cells (2n → 2n)
- Meiosis = Two divisions → 4 daughter cells (2n → n)
- Regulated by Cyclins, CDKs & Checkpoints
- Tumour suppressor genes = Brakes
- Oncogenes = Accelerators
- Imbalance causes Cancer



Understand the Cycle, Control the Future!

